

Lecture - 2

CO4: Compare various statistical methods of study data samples

CO5: Analyze and evaluation of different sets of data using hypothesis testing.

BIVARIATE DISCRETE RANDOM VARIABLES

The concept of single-dimensional random variable has been studied in detail. Proceeding in analogy with the one-dimensional case, concept of two-dimensional discrete random variables is discussed in this unit.

A situation where two-dimensional discrete random variable needs to be studied has already been given. To describe such situations mathematically, the study of two random variables is introduced.

Definition: Let X and Y be two discrete random variables defined on the sample space S of a random experiment then the function (X, Y) defined on the same sample space is called a two-dimensional discrete random variable. In other words, (X, Y) is a two-dimensional random variable if the possible values of (X, Y) are finite or countably infinite. Here, each value of X and Y is represented as a point (x, y) in the xy -plane.

As an illustration, let us consider the following example:

Let three balls b_1 , b_2 , b_3 be placed randomly in three cells. The possible outcomes of placing the three balls in three cells are shown in Table.

Arrangement No.	Placement of the Balls in		
	Cell 1	Cell 2	Cell 3
1	b_1	b_2	b_3
2	b_1	b_3	b_2
3	b_2	b_1	b_3
4	b_2	b_3	b_1
5	b_3	b_1	b_2
6	b_3	b_2	b_1
7	b_1, b_2	b_3	-
8	b_1, b_2	-	b_3
9	-	b_1, b_2	b_3
10	b_1, b_3	b_2	-
11	b_1, b_3	-	b_2
12	-	b_1, b_3	b_2
13	b_2, b_3	b_1	-
14	b_2, b_3	-	b_1
15	-	b_2, b_3	b_1
16	b_1	b_2, b_3	-
17	b_1	-	b_2, b_3
18	-	b_1	b_2, b_3
19	b_2	b_3, b_1	-
20	b_2	-	b_3, b_1
21	-	b_2	b_3, b_1
22	b_3	b_1, b_2	
23	b_3	-	b_1, b_2
24	-	b_3	b_1, b_2
25	b_1, b_2, b_3	-	-
26	-	b_1, b_2, b_3	-
27	-	-	b_1, b_2, b_3

Table: Possible Outcomes of Placing the Three Balls in Three Cells

Now, let X denote the number of balls in Cell 1 and Y be the number of cells occupied. Notice that X and Y are discrete random variables where X take on the values 0, 1, 2, 3 (number of balls in Cell 1 may be 0 or 1 or 2 or 3) and Y take on the values 1, 2, 3 (number of occupied cells may be 1 or 2 or 3). The possible values of two-dimensional random variable (X, Y) , therefore, are all ordered pairs of the values x and y of X and Y , respectively, i.e. are (0, 1), (0, 2), (0, 3), (1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (2, 3), (3, 1), (3, 2), (3, 3). Now, to each possible value (x_i, y_j) of (X, Y) , we can associate a number $p(x_i, y_j)$ representing $P(X = x_i, Y = y_j)$ as discussed.